

# Momentum and Collisions

SS ILP Physics (Intermediate), Week 5 Class 3



# Outline

- Elastic and inelastic Collisions
- Inelastic Collisions
- Completely Inelastic Collisions
- Elastic Collisions

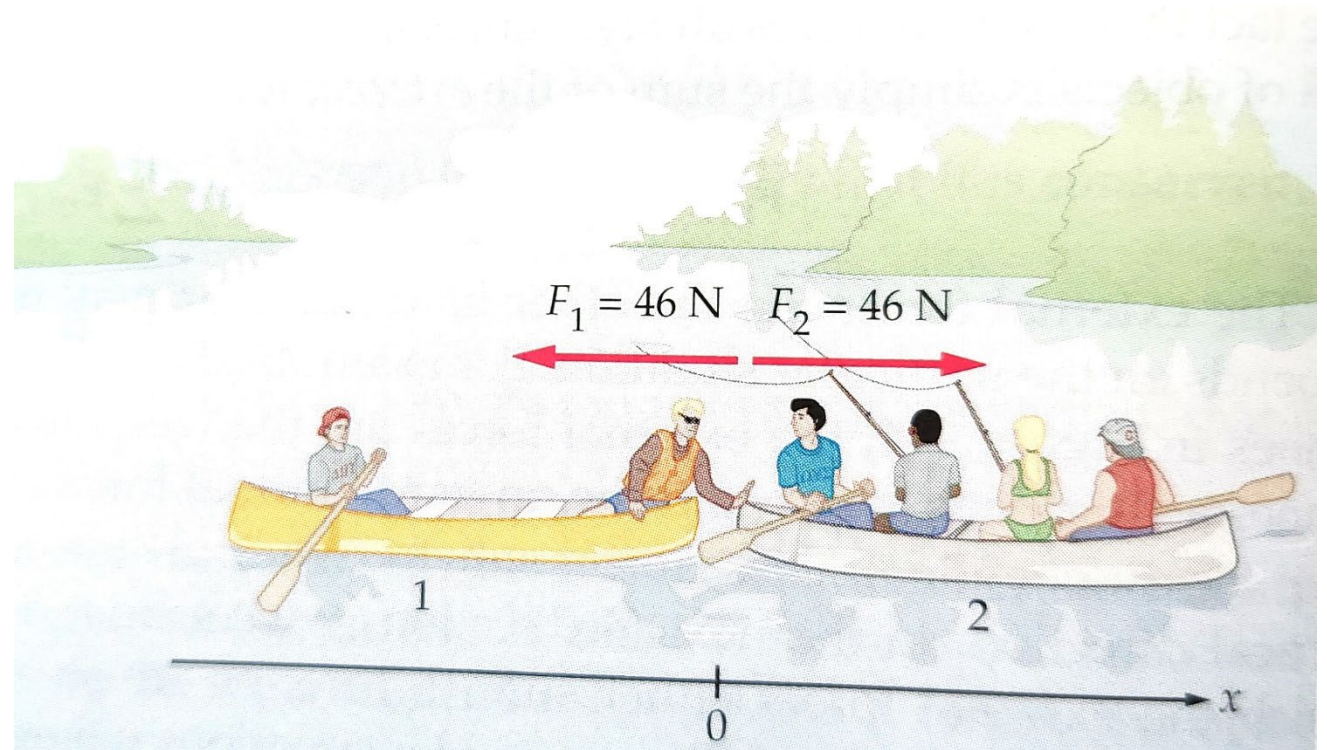
Reference:

Physics with Applications by Douglas Giancoli (7<sup>th</sup> Edition)

Physics by J. S. Walker

## Concept Question 0:

In Week 5 Class 2 slide-17 example, the final momentum of the system is equal to the initial momentum of the system. Is the final kinetic energy (a) equal to , (b) less than, or (c) greater than the initial kinetic energy?



J. S. Walker, Physics, Conceptual Checkpoint 9-2, p252

## Concept Question 0: Solution

In Week 5 Class 2 slide-17 example, the final momentum of the system is equal to the initial momentum of the system. Is the final kinetic energy (a) equal to , (b) less than, or (c) greater than the initial kinetic energy?

Initial kinetic energy = 0

$$\text{Final kinetic energy} = \frac{1}{2} m_1 v_{1,x}^2 + \frac{1}{2} m_2 v_{2,x}^2$$

# Elastic and Inelastic Collisions

In considering momentum conservation, we looked at what happens after there is a collision between two objects. Now, we are going to elaborate on the nature of these collisions.

**Elastic Collision:** A collision in which **kinetic energy is conserved**. When the satellite and the UFO **bounced off each other**, they experienced a **perfectly elastic collision**. If kinetic energy is lost to heat or anything else during the collision, it is called an inelastic collision.

**Inelastic Collision:** A collision in which **kinetic energy is not conserved**. When the satellite and the UFO **stick to each other**, they experienced a **perfectly inelastic collision**.

## Example:

You drop a 14-g ball from a height of 1.5 m and it only bounces back to a height of 0.85 m. What was the total impulse on the ball when it hit the floor? (Ignore air resistance.) Characterize the nature of the collision of the ball with the floor in this instance. Assume that the ball starts at rest prior to being dropped.

## Example: Solution

The impulse on the ball is its change in momentum. Call upward the positive direction, so that the final velocity is positive and the initial velocity is negative.

The speeds immediately before and immediately after the collision can be found from conservation of energy.

Take the floor to be the zero level for gravitational potential energy.

## Example: Solution

We split the solution into two parts:

(a) falling ball

(b) rising ball

(a) Since we are told that there is no air resistance, the kinetic energy of the ball at the bottom must be equal to the gravitational potential energy of the ball at the top.

## Example: Solution

As the ball hits the floor, the speed at the bottom must be negative since we stated that the positive direction is upwards.

$$\frac{1}{2}mv_{down}^2 = mgh_{down} \quad \rightarrow \quad v_{down} = -\sqrt{2gh_{down}}$$

(b) As the ball bounces, it rises up and we again must have the kinetic energy of the ball at the bottom must be equal to the gravitational potential energy of the ball at the top.

## Example: Solution

$$\frac{1}{2}mv_{up}^2 = mgh_{up} \rightarrow v_{up} = \sqrt{2gh_{up}}$$

Note that the sign of  $v_{up}$  is positive as the ball is bouncing UP.

So, the impulse is given by

$$\begin{aligned}\Delta p &= mv_{up} - mv_{down} = 0.014(\sqrt{2gh_{up}} - [-\sqrt{2gh_{down}}]) = \\ &0.014(\sqrt{2(9.8)(1.5)} + \sqrt{2(9.8)(0.85)}) = 0.13 \text{ kg m/s}\end{aligned}$$

## Example: Solution

Since the impulse is positive, the complete specification for impulse is that it is  $0.13 \text{ kg m/s}$  upwards.

Lastly, the kinetic energy is not the same before and after hitting the floor. So, kinetic energy is not conserved and the collision is inelastic. **If the collision was elastic, then ball would return to the originally dropped height.**

# Inelastic Collisions

In an inelastic collision, the momentum of a system is conserved,

$$\vec{P}_f = \vec{P}_i$$

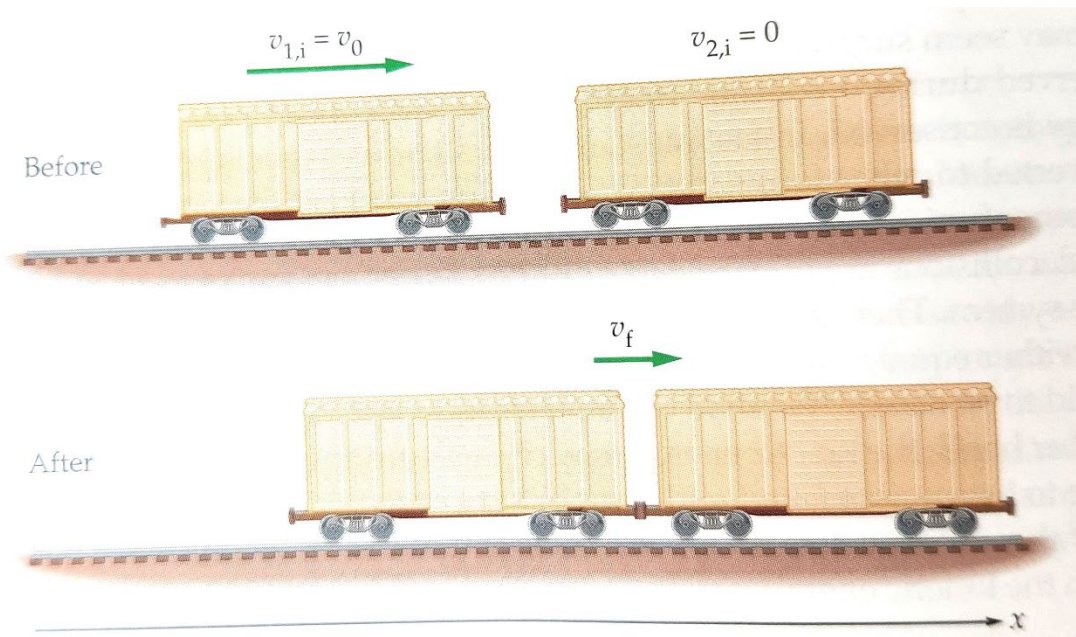
but its kinetic energy is not,

$$K_f \neq K_i$$

In the special case where objects stick together after the collision, we say that the collision is **completely inelastic**.

# Completely Inelastic Collisions

Consider a system of two train cars of mass  $m_1$  and  $m_2$  on a smooth, level track. Car 2 is at rest initially, and Car 1 moves toward it with a speed of  $v_0$ , as shown. When the cars collide, the coupling mechanism latches, causing the cars to stick together and move as a unit. What is the speed of the cars after collision?



J. S. Walker, Physics, p254

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_f$$

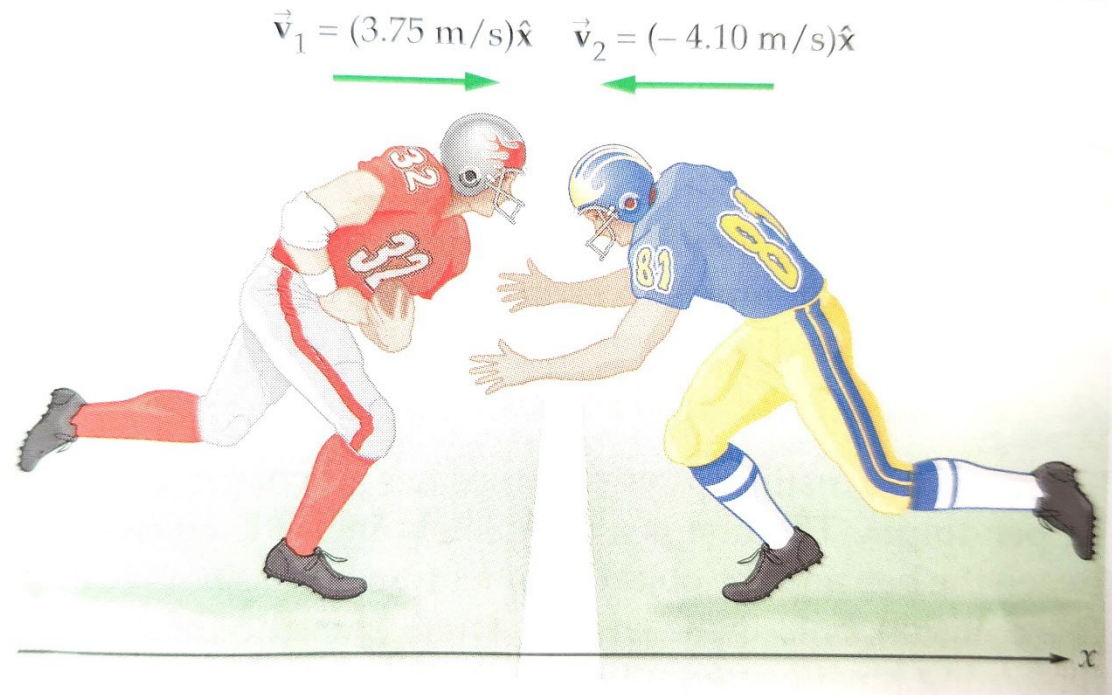
$$\vec{v}_f = \frac{m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i}}{(m_1 + m_2)}$$

$$v_f = \frac{m_1}{(m_1 + m_2)} v_0$$

## Example

On a touchdown attempt, a 95.0-kg running back runs toward the end zone at 3.75 m/s. A 111-kg linebacker moving at 4.10 m/s meets the runner in a head-on collision. If the two players stick together,

- (a) what is their velocity immediately after the collision?
- (b) What are the initial and final kinetic energies of the system?



J. S. Walker, Physics, Example 9-4, p255

## Example: Solution

On a touchdown attempt, a 95.0-kg running back runs toward the end zone at 3.75 m/s. A 111-kg linebacker moving at 4.10 m/s meets the runner in a head-on collision. If the two players stick together, (a) what is their velocity immediately after the collision?

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_f$$

$$\vec{v}_f = \frac{m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i}}{(m_1 + m_2)}$$

$$v_f = \frac{(95 \text{ kg})(3.75 \text{ m/s}) + (111 \text{ kg})(-4.10 \text{ m/s})}{(95 \text{ kg} + 111 \text{ kg})} = -0.48 \text{ m/s}$$

$$\vec{v}_f = (-0.48 \text{ m/s})\hat{i}$$

J. S. Walker, Physics, Example 9-4, p255

## Example: Solution

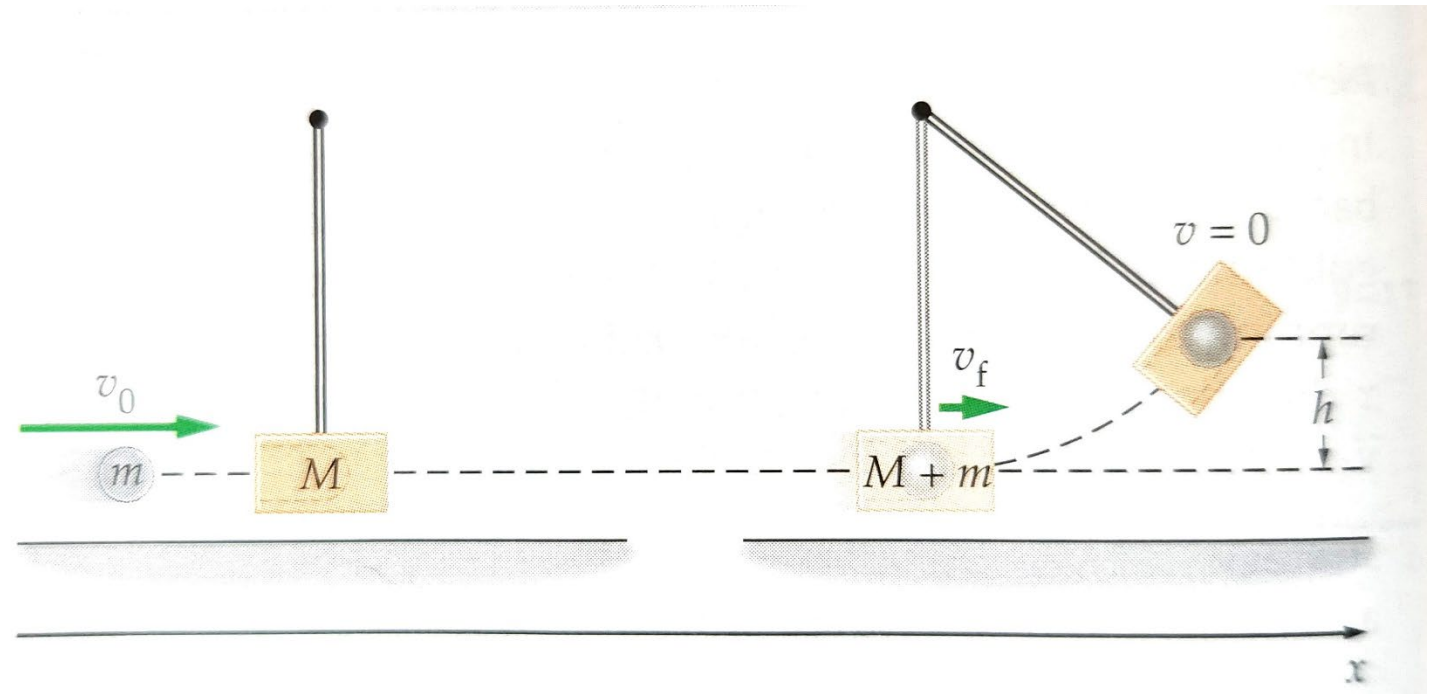
On a touchdown attempt, a 95.0-kg running back runs toward the end zone at 3.75 m/s. A 111-kg linebacker moving at 4.10 m/s meets the runner in a head-on collision. If the two players stick together,  
(b) What are the initial and final kinetic energies of the system?

$$K_i = \frac{1}{2}m_1v_{1,i}^2 + \frac{1}{2}m_2v_{2,i}^2 = \frac{1}{2}(95 \text{ kg})(3.75 \text{ m/s})^2 + \frac{1}{2}(111 \text{ kg})(-4.10 \text{ m/s})^2 = 1600 \text{ J}$$

$$K_f = \frac{1}{2}m_1v_{1,f}^2 + \frac{1}{2}m_2v_{2,f}^2 = \frac{1}{2}(95 \text{ kg})(-0.48 \text{ m/s})^2 + \frac{1}{2}(111 \text{ kg})(-0.48 \text{ m/s})^2 = 23.7 \text{ J}$$

## Case Problem 1

In a ballistic pendulum, an object of mass  $m$  is fired with an initial speed  $v_0$  at the bob of a pendulum. The bob has a mass  $M$ , and is suspended by a rod of negligible mass. After the collision, the object and the bob stick together and swing through an arc, eventually gaining a height  $h$ . Find the height  $h$  in terms of  $m$ ,  $M$ ,  $v_0$ , and  $g$ .



J. S. Walker, Physics, Example 9-5, p256

## Case Problem 1: Solution

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_f$$

$$v_f = \frac{mv_0 + M \times 0}{(m + M)} = \frac{m}{m + M} v_0$$

$$K_f = \frac{1}{2} m_1 v_{1,f}^2 + \frac{1}{2} m_2 v_{2,f}^2 = \frac{1}{2} m v_f^2 + \frac{1}{2} M v_f^2 = \frac{1}{2} m v_0^2 \left( \frac{m}{m + M} \right)$$

$$\frac{1}{2} m v_0^2 \left( \frac{m}{m + M} \right) = (m + M) g h \rightarrow h = \left( \frac{m}{m + M} \right)^2 \left( \frac{v_0^2}{2g} \right)$$

# Elastic Collisions

In an elastic collision, the momentum of a system is conserved,

$$\vec{P}_f = \vec{P}_i$$

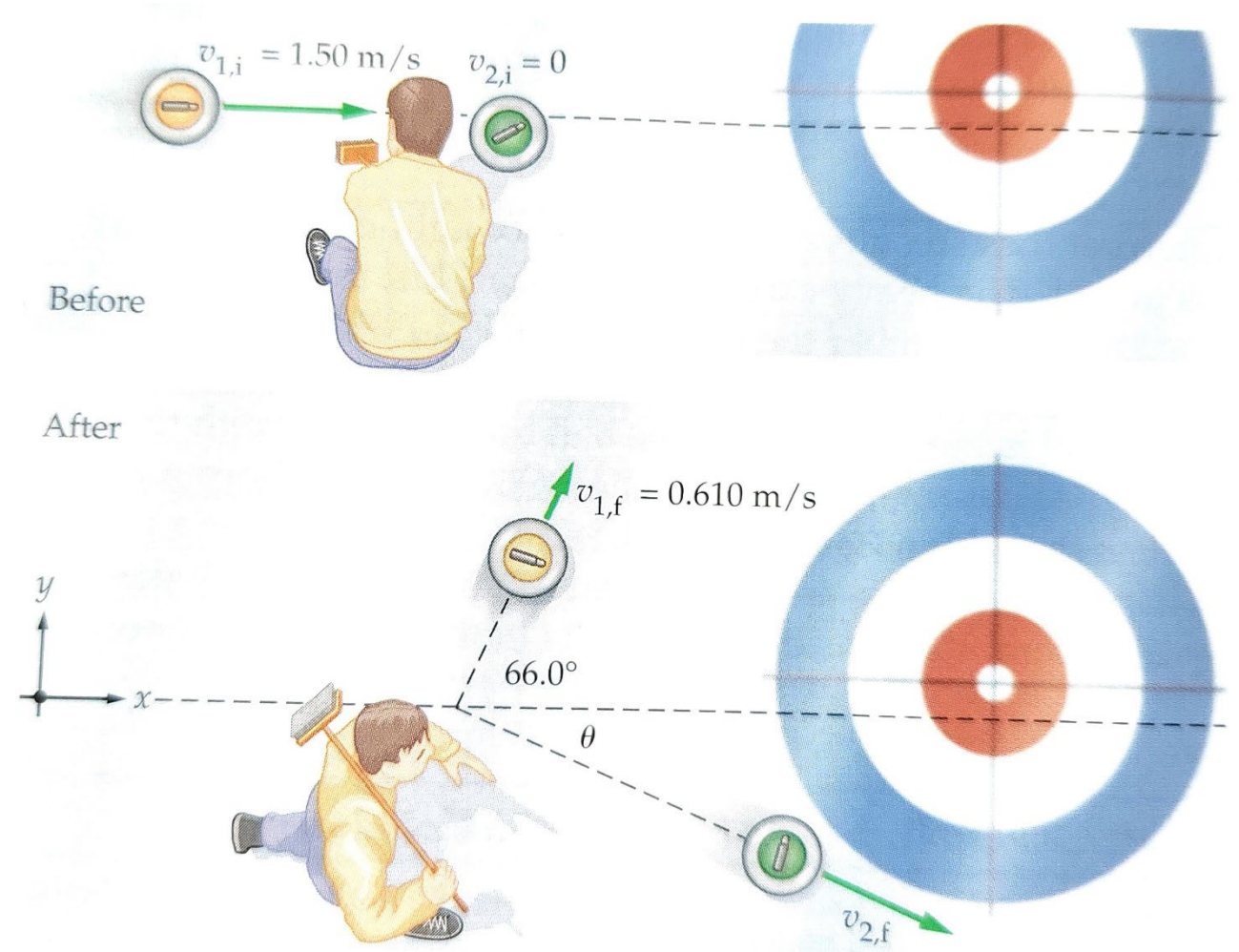
and its kinetic energy is conserved too,

$$K_f = K_i$$

Most collisions in everyday life are rather poor approximations to being elastic. However, the collision of objects that bounce off one another with little deformation provides a good approximation to an elastic collision. In subatomic world, elastic collisions are common.

## Example

Consider the collision of two 7.00-kg curling stones. Stone 2 is at rest initially, stone 1 approaches with a speed  $v_{1,i} = 1.50 \text{ m/s}$ . The collision is not head-on, and after the collision, stone 1 moves with a speed  $v_{1,f} = 0.610 \text{ m/s}$  in a direction  $66.0^\circ$  away from the initial line of motion. What is the speed and direction of stone 2?



## Example: Solution

$$\frac{1}{2}m_1v_{1,i}^2 = \frac{1}{2}m_1v_{1,f}^2 + \frac{1}{2}m_2v_{2,f}^2 \rightarrow v_{2,f} = 1.37 \text{ m/s}$$

$$m_1\vec{v}_{1,i} = m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f}$$

$$\hat{i}: m_1v_{1,i} = m_1v_{1,f} \cos 66^\circ + m_2v_{2,f} \cos \theta$$

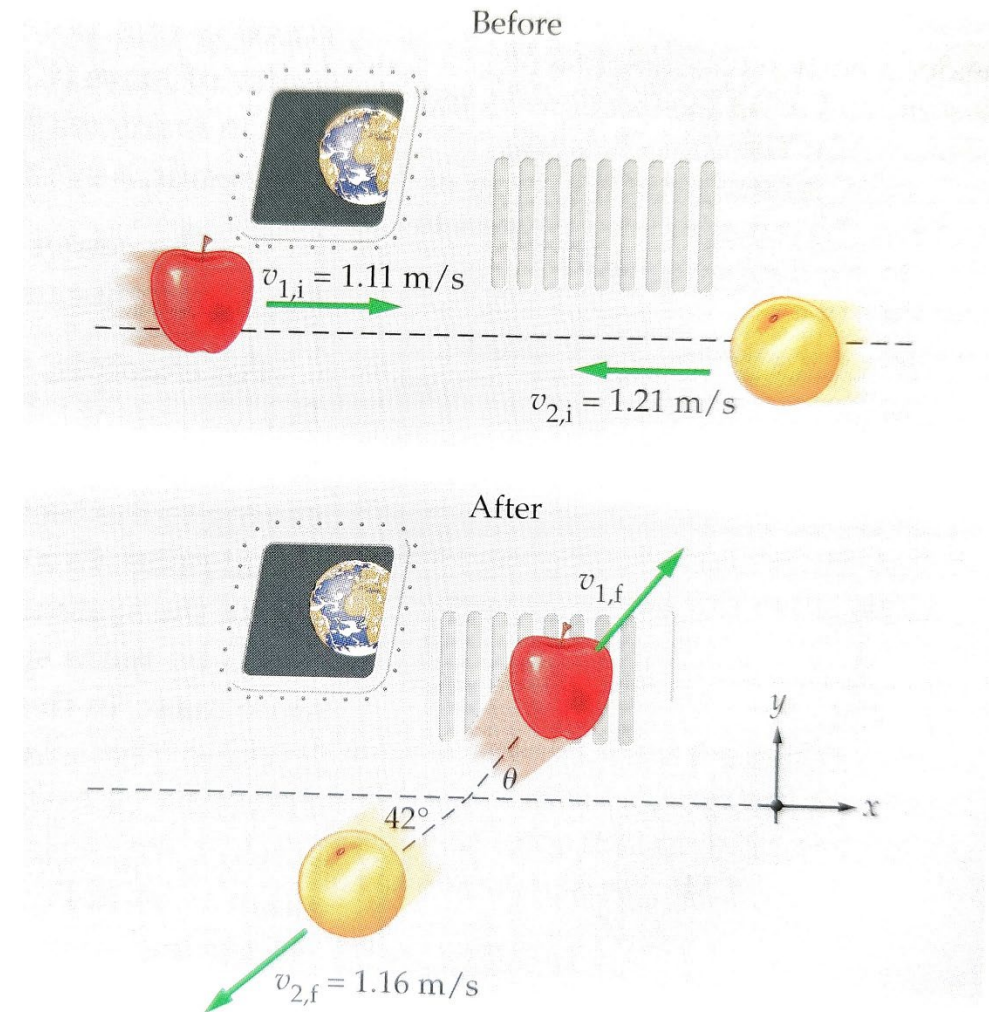
$$\hat{j}: m_1 \times 0 = m_1v_{1,f} \sin 66^\circ - m_2v_{2,f} \sin \theta \rightarrow \sin \theta = \frac{m_1v_{1,f} \sin 66^\circ}{m_2v_{2,f}}$$

$$\rightarrow \theta = 24^\circ$$

## Case Problem 2

Two astronauts on opposite ends of a spaceship are comparing lunches. One has an apple, the other has an orange. They decide to trade. Astronaut 1 tosses the 0.13-kg apple toward astronaut 2 with a speed of 1.11 m/s. The 0.16-kg orange is tossed from astronaut 2 to astronaut 1 with a speed of 1.21 m/s. Unfortunately, the fruits collide, sending the orange off with a speed of 1.16 m/s at an angle of  $42.0^\circ$  with respect to its original direction of motion. Find the final speed and direction of the apple, assuming an elastic collision. Give the apple's direction relative to its original direction of motion.

J. S. Walker, Physics, Example 9-7, p263



## Case Problem 2: Solution

$$\frac{1}{2}m_1v_{1,i}^2 + \frac{1}{2}m_2v_{2,i}^2 = \frac{1}{2}m_1v_{1,f}^2 + \frac{1}{2}m_2v_{2,f}^2 \rightarrow v_{1,f} = 1.17 \text{ m/s}$$

$$m_1\vec{v}_{1,i} + m_2\vec{v}_{2,i} = m_1\vec{v}_{1,f} + m_2\vec{v}_{2,f}$$

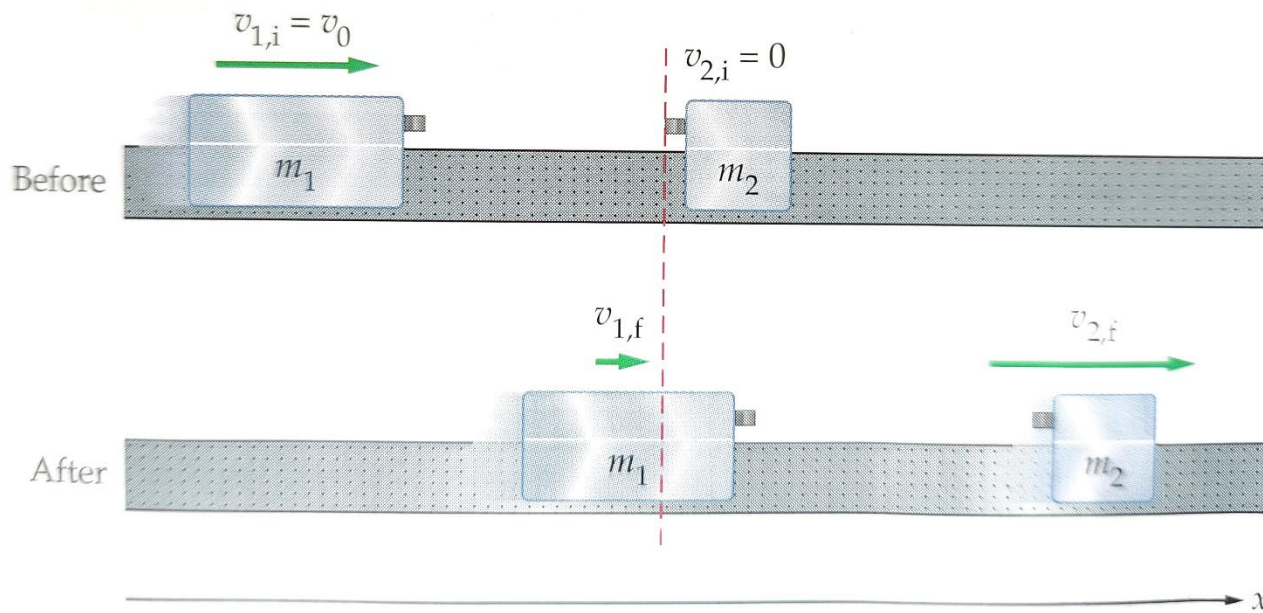
$$\hat{i}: m_1v_{1,i} - m_2v_{2,i} = m_1v_{1,f} \cos \theta - m_2v_{2,f} \cos 42^\circ$$

$$\hat{j}: 0 = m_1v_{1,f} \sin \theta - m_2v_{2,f} \sin 42^\circ \rightarrow \sin \theta = \frac{m_2v_{2,f} \sin 42^\circ}{m_1v_{1,f}} = 0.817$$

$$\rightarrow \theta = 54.7^\circ$$

# Elastic Collisions

Consider a head-on collision of two carts of mass  $m_1$  and  $m_2$  on an air track. The carts are provided with bumpers that give an elastic bounce when the carts collide. Suppose that initially cart 1 is moving to the right with a speed  $v_0$  toward cart 2, which is at rest. We have



J. S. Walker, Physics, p259

$$m_1 v_0 = m_1 v_{1,f} + m_2 v_{2,f}$$

$$\frac{1}{2} m_1 v_0^2 = \frac{1}{2} m_1 v_{1,f}^2 + \frac{1}{2} m_2 v_{2,f}^2$$



$$v_{1,f} = \left( \frac{m_1 - m_2}{m_1 + m_2} \right) v_0$$

$$v_{2,f} = \left( \frac{2m_1}{m_1 + m_2} \right) v_0$$

## Case Problem 3

A hoverfly is happily maintaining a fixed position about 10 ft above the ground when an elephant charges out of the bush and collides with it. The fly bounces elastically off the forehead of the elephant. If the initial speed of the elephant is  $v_0$ , is the speed of the fly after the collision roughly equal to (a)  $v_0$ , (b)  $1.5v_0$ , or (c)  $2v_0$ ?

## Case Problem 3: Solution

A hoverfly is happily maintaining a fixed position about 10 ft above the ground when an elephant charges out of the bush and collides with it. The fly bounces elastically off the forehead of the elephant. If the initial speed of the elephant is  $v_0$ , is the speed of the fly after the collision roughly equal to (a)  $v_0$ , (b)  $1.5v_0$ , or (c)  $2v_0$ ?

$$v_{1,f} = \left( \frac{m_1 - m_2}{m_1 + m_2} \right) v_0 \approx v_0$$
$$v_{2,f} = \left( \frac{2m_1}{m_1 + m_2} \right) v_0 \approx 2v_0$$

# Off-class practice question(s)

## Example:

A sphere of mass 1.5 kg and speed 2 m/s collides head-on with an identical sphere of mass 0.5 kg and speed 1 m/s.

As a result of the impact, the direction of motion of the lighter sphere is reversed and its speed becomes 3.5 m/s.

Determine: (a) the speed of the heavier sphere after impact, (b) the loss in kinetic energy due to the impact and (c) the nature of the collision.

## Example: Solution

(a) The first fact to note is the term **head-on** which means that the two spheres are approaching each other from **opposite directions**.

Next, we set the direction of motion of the heavier sphere (since it maintains consistent direction of motion) to be positive. Applying the conservation of momentum principle, we have  $1.5(2) + 0.5(-1) = 1.5(v) + 0.5(3.5)$

→  $v = 0.5 \text{ m/s}$  . Note that the post-collision speed of the lighter sphere has changed sign.

## Example: Solution

$$\text{(b) Total Kinetic Energy (pre-collision)} = \frac{1}{2} (1.5)(2)^2 + \frac{1}{2} (0.5)(-1)^2 = 3.25 \text{ kg m}^2/\text{s}^2$$

$$\text{Total Kinetic Energy (post-collision)} = \frac{1}{2} (1.5)(0.5)^2 + \frac{1}{2} (0.5)(3.5)^2 = 3.25 \text{ kg m}^2/\text{s}^2$$

Since the total kinetic energy before and after collision is the same, there is no loss of kinetic energy.

## Example: Solution

(c) In view of the answer in (b), the collision is elastic since there was no loss of kinetic energy.

## Case Problem:

A 1200-kg car moving at 2.5 m/s is struck in the rear by a 2600-kg truck moving at 6.2 m/s. if the vehicles stick together after the collision, what is their speed immediately after colliding? (Assume that external forces may be ignored.)

## Case Problem: Solution

A 1200-kg car moving at 2.5 m/s is struck in the rear by a 2600-kg truck moving at 6.2 m/s. if the vehicles stick together after the collision, what is their speed immediately after colliding? (Assume that external forces may be ignored.)

$$m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i} = (m_1 + m_2) \vec{v}_f$$

$$\vec{v}_f = \frac{m_1 \vec{v}_{1,i} + m_2 \vec{v}_{2,i}}{(m_1 + m_2)}$$

$$v_f = \frac{(1200 \text{ kg})(2.5 \text{ m/s}) + (2600 \text{ kg})(6.2 \text{ m/s})}{(1200 \text{ kg} + 2600 \text{ kg})} = 5.0 \text{ m/s}$$